

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Cable Sizing Calculations
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003
Document Revision	R01
Document Status	Issued For Review
Originator	Noel Kari
Security Classification	Restricted
Issue Date	27 Jul 2022

Cable Sizing Calculation Report

R01	27.07.22	Issued for Review	N. Kari	V. Ani	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	27.07.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

This document is made available subject to the condition that the recipient will neither use nor disclose the contents except as agreed in writing with the copyright owner. Copyright is vested in Brass Fertilizer and Petrochemical Company Limited (BFPCL) © All rights reserved.

Neither the whole nor any part of this document may be reproduced or distributed in any form or by any means (electronic, mechanical, reprographic, recording, or otherwise) without the prior written consent of the copyright owner.

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited		
Project:	Gas Processing & Methanol Complex	Project No. 11998A	
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003	

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

[illegible]

Table of Contents

1. NOTES 1	5
2. NOTES 2	6
3. BBOU LOW VOLTAGE CABLE	7
4. ELEPA LOW VOLTAGE CABLE	8
5. AKABEL LOW VOLTAGE CABLE	9
6. REACTANCE CALCULATION	10

	Document No.	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003	REV	R01	
	Doc. Title	Cable Sizing Calculations Report			

CABLE SIZING

OBJECTIVE:

To determine a suitable cable size for electrical equipments required for Gas Gathering Manifold Station

METHODOLOGY:

To determine a suitable cable size for electrical equipment the following criteria must be considered and are performed in the following order -

a) Current-carrying capacity.

To calculate the current-carrying capacity requirements of a circuit:

i) Determine the current requirements of the circuit.

ii) Determine the method of cable installation to be used (primary de-rating factor), ie. proximity to other loaded cables, method of laying and ventilation available etc.

iii) Determine the environmental conditions in the vicinity of the cable installation (secondary de-rating factor), ie. ambient air/soil temperature, depth of laying etc. See table table A.52-12 to 15 of IEC 60364-5-52

Ambient Temperature @ 40 degree Celcius = 0.83

To select the appropriate sized cable for the circuit, multiply the value of current determined in i) by the de-rating factors determined in ii) and iii).

The resulting value of current represents the minimum current-carrying capacity required of the circuit. Refer to the tables of current-carrying capacity for the different cable types, table A.52-1 of IEC 60364-5-52

Taking into account the method of installation employed, the smallest conductor size which has a tabulated current-carrying capacity equal to or in excess of the previously calculated minimum value, will be considered to be the minimum cable size satisfying the first of the four criteria.

$$V_{d3\phi} = \frac{I \times L \times \sqrt{3}(R_c \cos \theta + X_c \sin \theta)}{1000}$$

b) Permissible Voltage Drop, with respect to Load Power Factor.

Voltage drop can be defined as the difference in magnitude of the voltage at the supply compared to the voltage at the load. This voltage drop can be attributed to a number of factors, such as the load's power factor, the cable's internal resistance and reactance, and the cable length. The three phase voltage drop permitted on the circuit run is calculated using the following equation:

where:

V_{d3φ} = Three phase voltage drop, volts (V) Power Factor (pf)= Cosθ

I = Current flowing in cable, amperes (A)

R_c = A.C resistance of cable, ohm/1000 meters (Ω/1000m)

X_c = Reactance of cable, ohm/1000 meters (Ω/1000m)

The single phase voltage drop permitted on the circuit run is calculated using the following equation,

NOTE:- For the purpose of this calculation; Cables are laid in cable tray and directly buried underground , hence derating shall be ambient temprature - 0.83, Installation at 0.6m depth - 0.98, Cable grouping - 0.55

	Document No.	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003	REV	R01	
	Doc. Title	Cable Sizing Calculations Report			

CABLE SIZING

$$V_{d1\phi} = \frac{I \times L \times 2(R_c \cos\theta + X_c \sin\theta)}{1000}$$

where:

$V_{d1\phi}$ = single phase voltage drop, volts (V) Power Factor (pf) = $\cos\theta$
 I = current flowing in cable, amperes (A)
 R_c = AC resistance of cable, ohm/1000 meters ($\Omega/1000m$)
 X_c = reactance of cable, ohm/1000 meters ($\Omega/1000m$)
 L = route length of circuit, meters (m)

The D.C. voltage drop permitted on the circuit run is calculated using the following equation:

$$V_{DC} = \frac{2ILR_c}{1000}$$

where:

V_{DC} = single phase voltage drop, volts (V)
 I = current flowing in cable, amperes (A)
 R_c = DC resistance of cable, ohm/1000 meters ($\Omega/1000m$)
 L = route length of circuit, meters (m)

The permissible voltage drop check is calculated on the cable size selected in step a) previously. If the calculated voltage drop is found to exceed the maximum permissible voltage drop specified, the selected cable size is increased to the next sized cable, until the calculated voltage drop is equal to or less than the maximum permissible voltage drop.

Note: Maximum permissible voltage drop between the point of supply and the load (e.g. motor) shall not exceed 5%. However connection of the load in question may be from a sub distribution board, therefore the maximum voltage drop for the calculation would be less than 5% to allow for a voltage drop in the initial circuit (point of supply) feeding the sub distribution board. See below;

Feeders (from main to sub-main Switchboard & MCC):	2%
Power Generator and LV Switchgaer:	2%
Motor Terminal Voltage (During motor starting voltage at motor terminals shall ensure a sufficient accelerating torque and voltage drop at bus bar shall not cause shutdown of other consumers.)	5% (15%max. @ starting)
Lighting sub-circuits and sockets	2% average with a maximum of 8% at to the farthest fitting.
For the socket outlet 30% is used for derating the ratings, since the load due come on intermittently.	
Other (heaters and packages)	5%

NOTE: The loads captured in this calculation sheet is based on loads contained in present revision of load list. this sizing calculation shall be updated once there are load changes on the load list.

			DOCUMENT NUMBER						EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003										REV		R01			
			DOCUMENT TITLE						Cable Sizing Calculations Report (BBOU)															
Ambient Temp. °C 43																								
Cable on Tray																								
Total																								
			Derating Factor 0.91						0.93						0.8463									
EQUIPMENT DESCRIPTION			VOLTAGE (V)	KW	P.F (COS Q)	FULL LOAD CURRENT (A)	BREAKER RATING (Min) (A)	SELECTED BREAKER RATING (A)	NEXANS CABLE CATALOGUE				CABLE IN GROUP	DERATING FACTOR	CABLE DERATING AMPACITY (A)	MIN. NO OF CABLE IN PARALLEL PER PHASE	NO. OF CABLE IN PARALLEL SELECETD (n)	SHORT CIRCUIT RATING FOR SELECTED CABLE (KA/Sec)	VOLTAGE DROP					
									CABLE SIZE (Sqmm)	CABLE SIZE (core)	CABLE RESISTANCE @ 90°C (Ohm/Km)	CABLE REACTANCE @ 50°C (Ohm/Km)							CABLE IMPEDANCE REACTANCE (Z)	CABLE AMPACITY (A)	CABLE LENGTH (MTR)	VOLTAGE DROP (mV/A/m)	% VOLTAGE DROP	ALLOWABLE VOLTAGE DROP (%)
P-SB01	TR01 (STEPDOWN TRANSFORMER)	SB01 (LV SWITCHBOARD)	400	55.4	0.8	99.95	124.94	120	50	4	0.494	0.0751	0.50	192	1	0.846	162.49	1	1	7.15	30	2.595	0.65	2
P-DB01	SB01 (LV SWITCHBOARD)	DB-01 (ELECTRICAL ROOM LIGHTING & SMALL POWER DB)	400	11	0.8	19.85	24.81	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	1.351	0.34	2
P-DB02	SB01 (LV SWITCHBOARD)	DB-02 (RTU ROOM LIGHTING AND SMALL POWER DB)	400	11	0.8	19.85	24.81	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	20	2.703	0.68	2
P-DB03	SB01 (LV SWITCHBOARD)	DB-03 (GUARD & OFFICE BUILDING LIGHTING & SMALL POWER DB)	400	17.7	0.8	31.93	39.92	40	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	20	2.579	0.64	2
P-DB04	SB01 (LV SWITCHBOARD)	DB-04 (EXTERNAL LIGHTING DB)	400	22.2	0.8	40.05	50.07	40	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	10	1.617	0.40	2
P-UPS01	SB01 (LV SWITCHBOARD)	UPS01	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-UPS02	SB01 (LV SWITCHBOARD)	UPS02	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-UPS03	SB01 (LV SWITCHBOARD)	UPS03 (UPS BY-PASS)	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-DB05	SB01 (LV SWITCHBOARD)	DB-05 (SPDC FEEDER)	400	3	0.8	5.41	6.77	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	42	1.548	0.39	5
P-TRF02	SB01 (LV SWITCHBOARD)	TRF-02 (CATHODIC PROTECTION TRANSFORMER)	400	5	0.8	9.02	11.28	32	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	10	0.364	0.09	5
P-PM01	SB01 (LV SWITCHBOARD)	PM-01 (CORROSION INHIBITOR MOTOR 1)	400	1	0.8	1.80	2.26	20	4	4	5.88	0.093	5.88	42	1	0.846	35.54	1	1	0.572	30	0.551	0.14	5
P-PM02	LV SWITCHBOARD (SB - 7403)	PM-02 (CORROSION INHIBITOR MOTOR 1)	400	1	0.8	1.80	2.26	20	4	4	5.88	0.093	5.88	42	1	0.846	35.54	1	1	0.572	30	0.551	0.14	5

			DOCUMENT NUMBER						EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003										REV		R01			
			DOCUMENT TITLE						Cable Sizing Calculations Report (ELEPA)															
Ambient Temp. °C 43																								
Cable on Tray																								
Total																								
			Derating Factor 0.91						0.93						0.8463									
EQUIPMENT DESCRIPTION			VOLTAGE (V)	KW	P.F (COS Q)	FULL LOAD CURRENT (A)	BREAKER RATING (Min) (A)	SELECTED BREAKER RATING (A)	NEXANS CABLE CATALOGUE				CABLE IN GROUP	DERATING FACTOR	CABLE DERATING AMPACITY (A)	MIN. NO OF CABLE IN PARALLEL PER PHASE	NO. OF CABLE IN PARALLEL SELECETD (n)	SHORT CIRCUIT RATING FOR SELECTED CABLE (KA/Sec)	VOLTAGE DROP					
									CABLE SIZE (Sqmm)	CABLE SIZE (core)	CABLE RESISTANCE @ 90°C (Ohm/Km)	CABLE REACTANCE @ 50°C (Ohm/Km)							CABLE IMPEDANCE REACTANCE (Z)	CABLE AMPACITY (A)	CABLE LENGTH (MTR)	VOLTAGE DROP (mV/A/m)	% VOLTAGE DROP	ALLOWABLE VOLTAGE DROP (%)
P-SB01	TR01 (STEPDOWN TRANSFORMER)	SB01 (LV SWITCHBOARD)	400	55.4	0.8	99.95	124.94	120	50	4	0.494	0.0751	0.50	192	1	0.846	162.49	1	1	7.15	30	2.595	0.65	2
P-DB01	SB01 (LV SWITCHBOARD)	DB-01 (ELECTRICAL ROOM LIGHTING & SMALL POWER DB)	400	11	0.8	19.85	24.81	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	1.351	0.34	2
P-DB02	SB01 (LV SWITCHBOARD)	DB-02 (RTU ROOM LIGHTING AND SMALL POWER DB)	400	11	0.8	19.85	24.81	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	20	2.703	0.68	2
P-DB03	SB01 (LV SWITCHBOARD)	DB-03 (GUARD & OFFICE BUILDING LIGHTING & SMALL POWER DB)	400	17.7	0.8	31.93	39.92	40	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	20	2.579	0.64	2
P-DB04	SB01 (LV SWITCHBOARD)	DB-04 (EXTERNAL LIGHTING DB)	400	22.2	0.8	40.05	50.07	40	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	10	1.617	0.40	2
P-UPS01	SB01 (LV SWITCHBOARD)	UPS01	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-UPS02	SB01 (LV SWITCHBOARD)	UPS02	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-UPS03	SB01 (LV SWITCHBOARD)	UPS03 (UPS BY-PASS)	400	4	0.8	7.22	9.02	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	10	0.491	0.12	5
P-DB05	SB01 (LV SWITCHBOARD)	DB-05 (SPDC FEEDER)	400	3	0.8	5.41	6.77	20	6	4	3.93	0.0887	3.93	54	1	0.846	45.70	1	1	0.858	42	1.548	0.39	5
P-TRF02	SB01 (LV SWITCHBOARD)	TRF-02 (CATHODIC PROTECTION TRANSFORMER)	400	5	0.8	9.02	11.28	32	10	4	2.33	0.084	2.33	75	1	0.846	63.47	1	1	1.43	10	0.364	0.09	5
P-PM01	SB01 (LV SWITCHBOARD)	PM-01 (CORROSION INHIBITOR MOTOR 1)	400	1	0.8	1.80	2.26	20	4	4	5.88	0.093	5.88	42	1	0.846	35.54	1	1	0.572	30	0.551	0.14	5
P-PM02	LV SWITCHBOARD (SB - 7403)	PM-02 (CORROSION INHIBITOR MOTOR 1)	400	1	0.8	1.80	2.26	20	4	4	5.88	0.093	5.88	42	1	0.846	35.54	1	1	0.572	30	0.551	0.14	5

[illegible]

ATTACHMENT-A
DATA FOR RESISTANCES AND REACTANCES

SIZE OF CABLE (SQ. MM)	RESISTANCE AT 20 ⁰ C (R @ 20) (Ohms/km)	RESISTANCE AT 90 ⁰ C (R @ 90) (Ohms/km)	REACTANCE AT 50Hz (X) (Ohms/km)
2.5	7.41	9.448	0.090
4	4.61	5.878	0.092
6	3.08	3.927	0.088
10	1.83	2.333	0.083
16	1.15	1.466	0.081
25	0.73	0.927	0.079
35	0.52	0.668	0.077
50	0.39	0.493	0.076
70	0.27	0.342	0.075
95	0.19	0.246	0.073
120	0.15	0.195	0.072
150	0.12	0.158	0.073
185	0.10	0.126	0.073
240	0.08	0.096	0.072
300	0.06	0.077	0.072
400	0.05	0.060	0.070